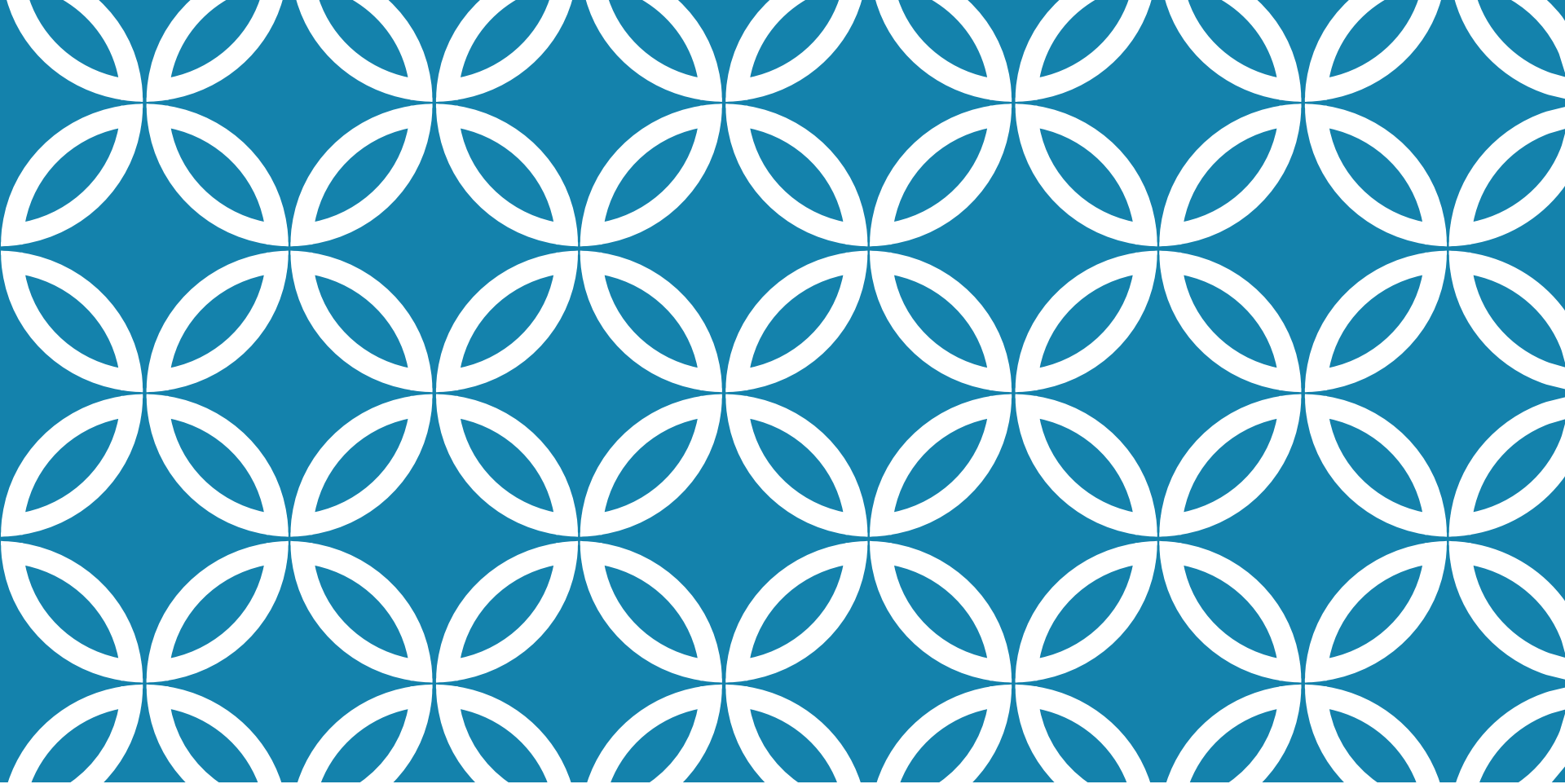




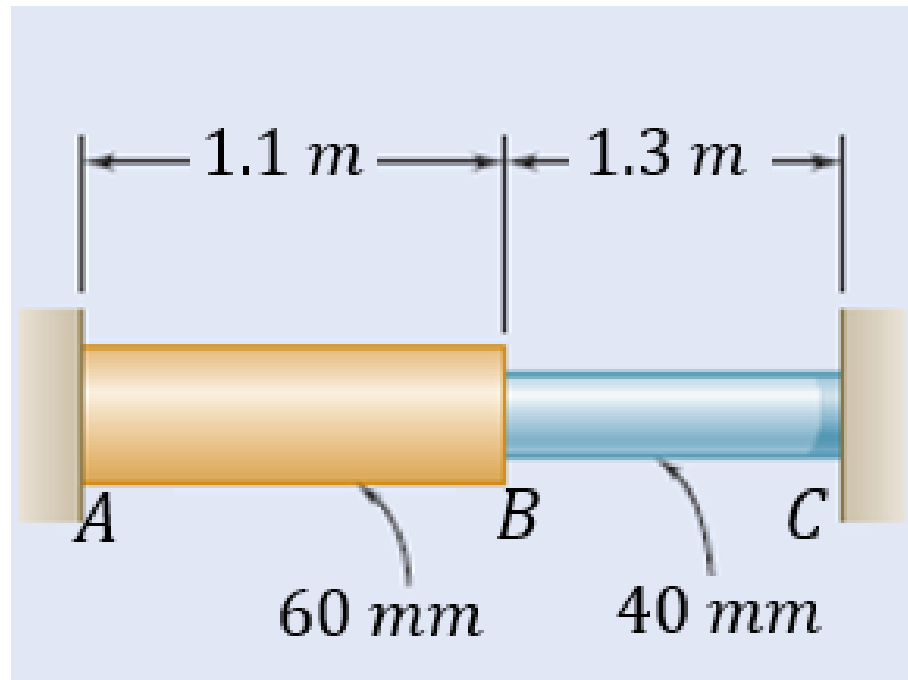
SESSION 1 GUIDE



UNIT SYSTEM

	SI		SI(mm)		US Unit
Length	m		mm		in
Force	N		N		1bf
Mass	kg		T (10^3 kg)		1bf s ² /in
Time	s		s		s
Stress	Pa		MPa		psi
Energy	J		mJ (10^{-3} J)		in 1bf
Density	Kg/m ³		t/mm ³		1bf s ² /in ⁴
Ace	m/s ²		mm/s ²		

EXERCISE 1: AXISYMMETRIC 2D



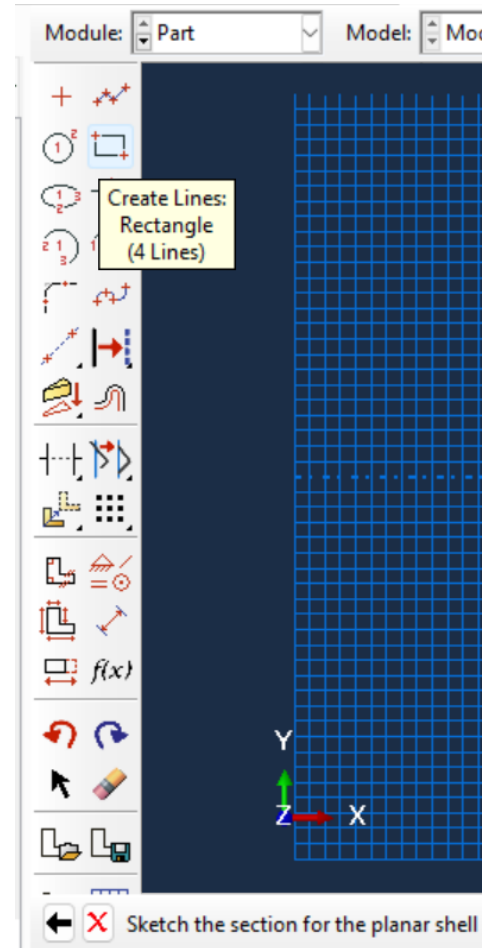
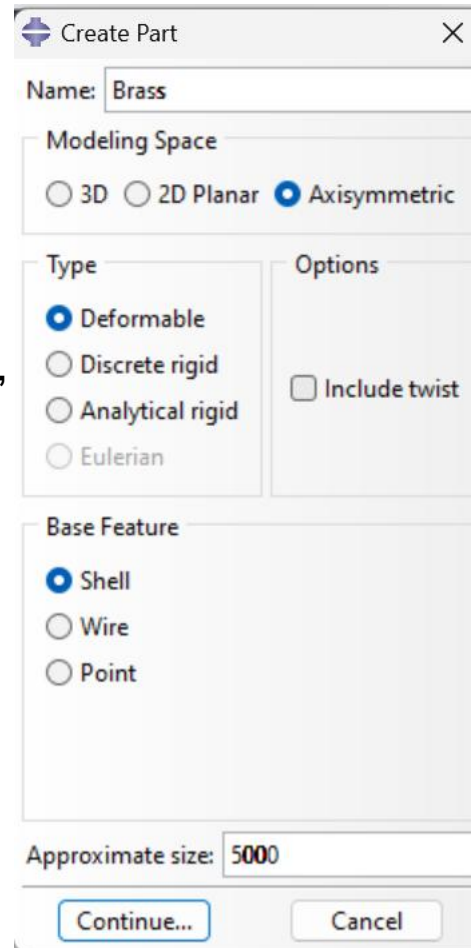
1. PART: BRASS AND ALUMINUM RODS

➤ Create Brass Part



➤ In the “Create Part” interface,

- Put the “Name” as “Brass”
- Tick the “Axisymmetric”
- Change the “Approximate size” to “5000”
- Click “Continue”
- Draw the rectangle in the grid
 - (Rectangle size: 30x1100)
- Click Done



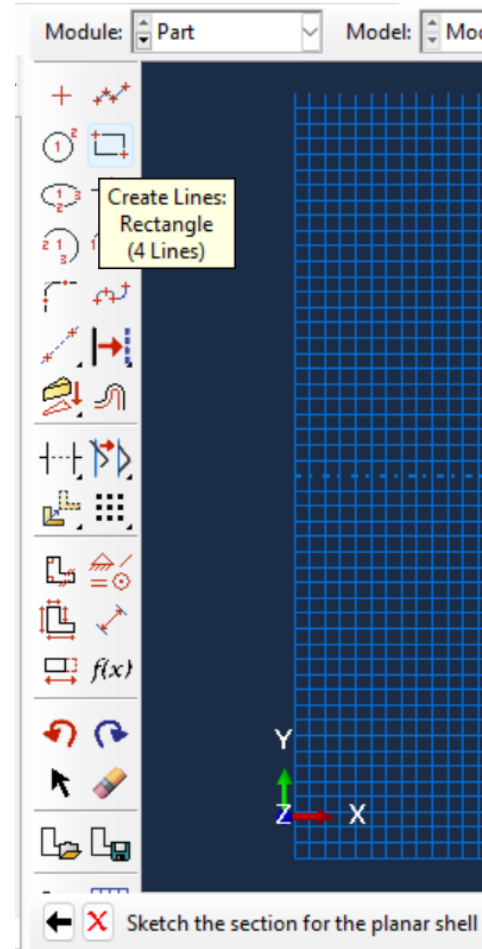
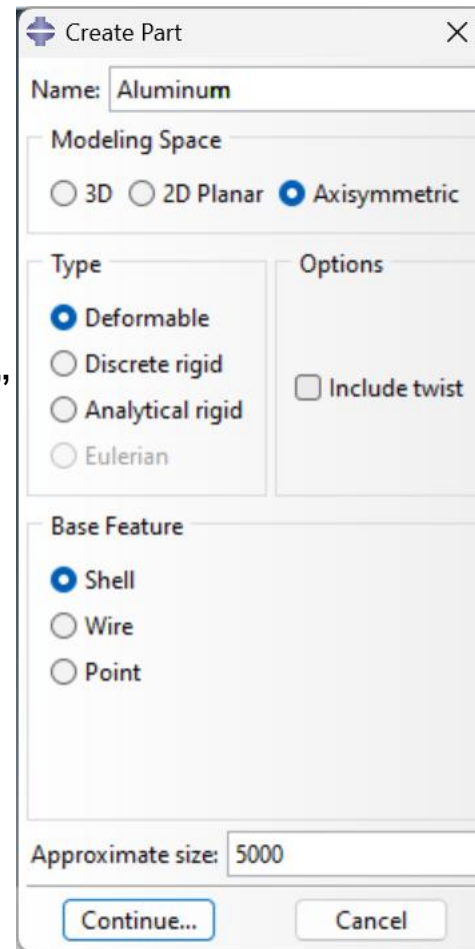
1. PART: BRASS AND ALUMINUM RODS

➤ Create Aluminum Part



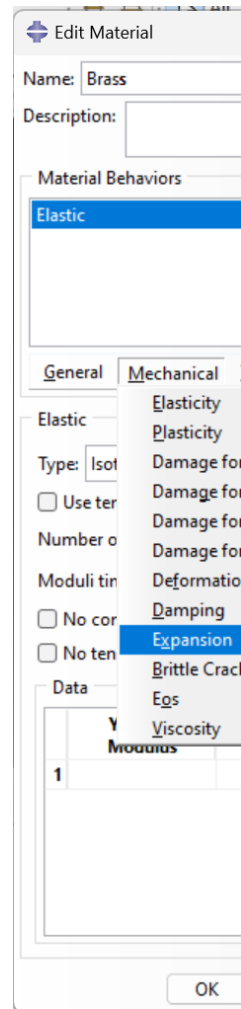
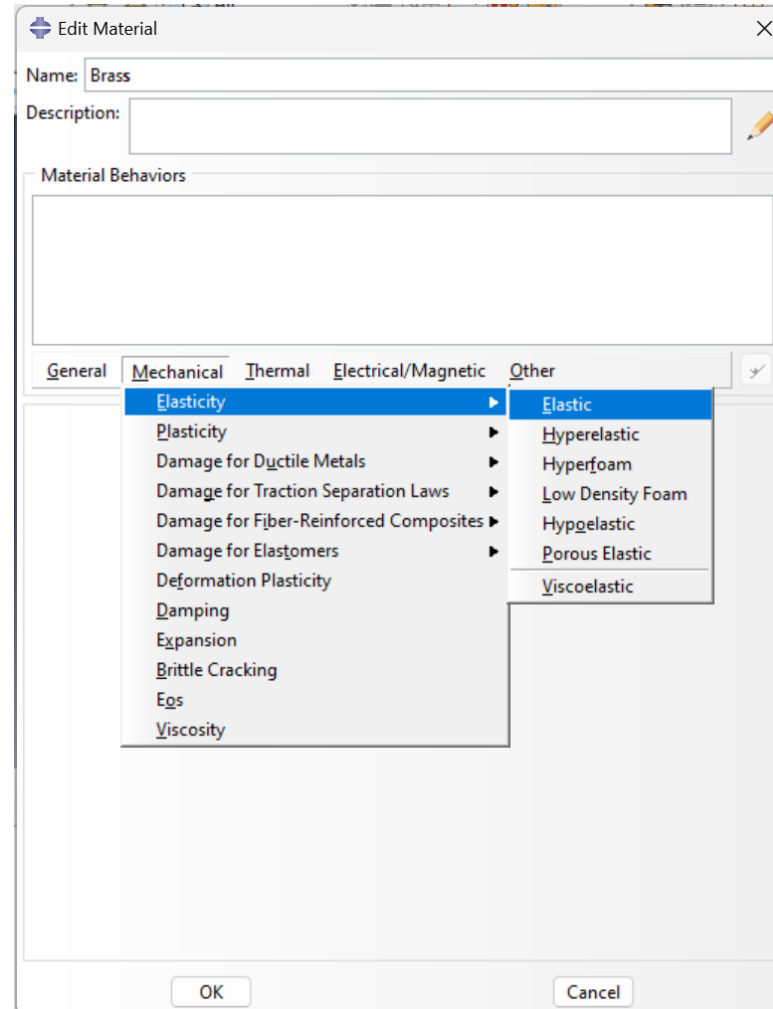
➤ In the “Create Part” interface,

- Put the “Name” as “Aluminum”
- Tick the “Axisymmetric”
- Change the “Approximate size” to “5000”
- Click “Continue”
- Draw the rectangle in the grid
 - (Rectangle size: 20x1300)
- Click Done



2. PROPERTY: ASSIGN PROPERTIES TO BRASS AND ALUMINUM PARTS

- Create Property
 - Input “Name” as Brass
 - Click on “Mechanical”, “Elasticity”, then “Elastic”. Input the values of “Young’s Modulus” and “Poisson’s Ratio”
 - Click on “Mechanical”, “Expansion”. Input the value of “Expansion Coeff alpha”
- Create Section
- Assign Section



2. PROPERTY: ASSIGN PROPERTIES TO BRASS AND ALUMINUM PARTS

- Create Property
 - Input “Name” as Aluminum
 - Click on “Mechanical”, “Elasticity”, then “Elastic”. Input the values of “Young’s Modulus” and “Poisson’s Ratio”
 - Click on “Mechanical”, “Expansion”. Input the value of “Expansion Coeff alpha”
- Create Section
- Assign Section

The screenshot shows the 'Edit Material' dialog box for Aluminum. The 'Name' field is set to 'Aluminum'. The 'Description' field is empty. The 'Material Behaviors' list shows 'Elastic' and 'Expansion' selected. The 'Elastic' tab is active, showing 'Type: Isotropic', 'Use temperature-dependent data' (unchecked), 'Number of field variables: 0', 'Moduli time scale (for viscoelasticity): Long-term', 'No compression' (unchecked), and 'No tension' (unchecked). The 'Data' table shows Young's Modulus as 72000 and Poisson's Ratio as 0.3.

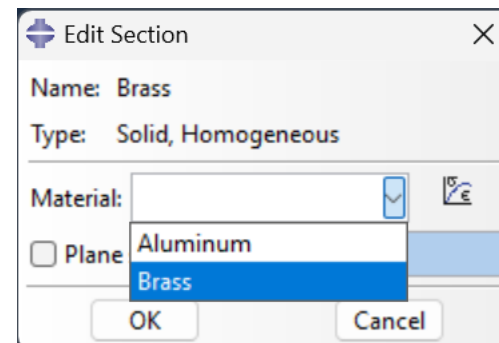
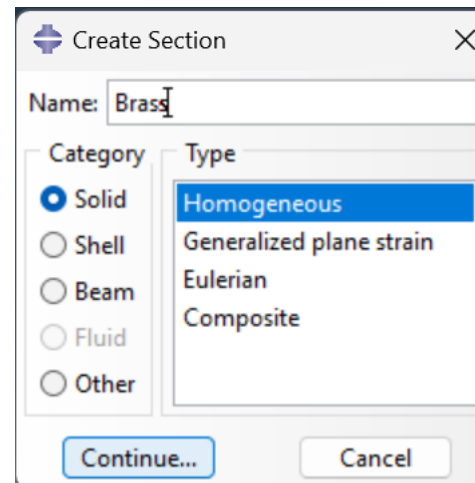
	Young's Modulus	Poisson's Ratio
1	72000	0.3

The screenshot shows the 'Edit Material' dialog box for Aluminum, with the 'Expansion' tab active. The 'Type' is set to 'Isotropic', 'Use user subroutine UEXPAN' (unchecked), 'Reference temperature: 0', 'Use temperature-dependent data' (unchecked), and 'Number of field variables: 0'. The 'Data' table shows 'Expansion Coeff alpha' as 2.39E-005.

	Expansion Coeff alpha
1	2.39E-005

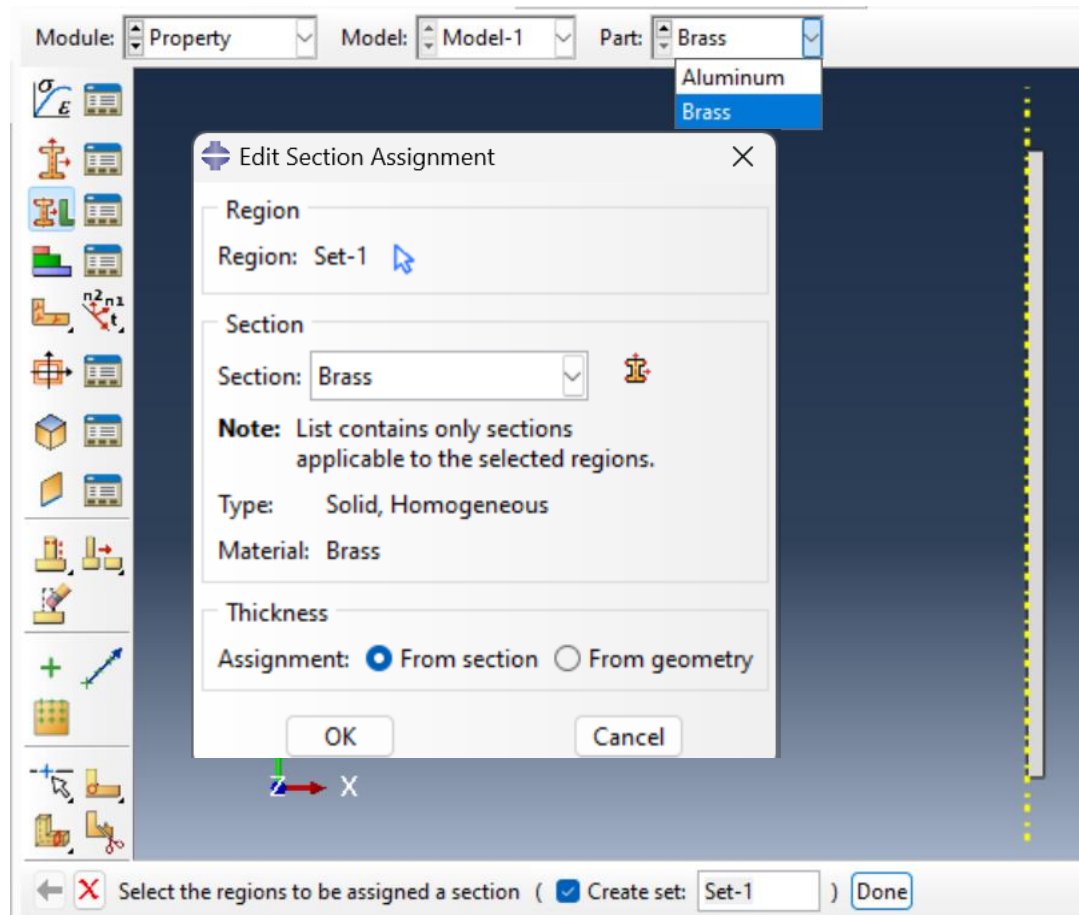
2. PROPERTY: ASSIGN PROPERTIES TO BRASS AND ALUMINUM PARTS

- Create Section for Brass
 - Input “Name” as Brass
 - Click Continue
 - Link the “Material” to “Brass”
 - Click “Ok”
- Create Section for Aluminum
 - Input “Name” as Aluminum
 - Click Continue
 - Link the “Material” to “Aluminum”
 - Click “Ok”



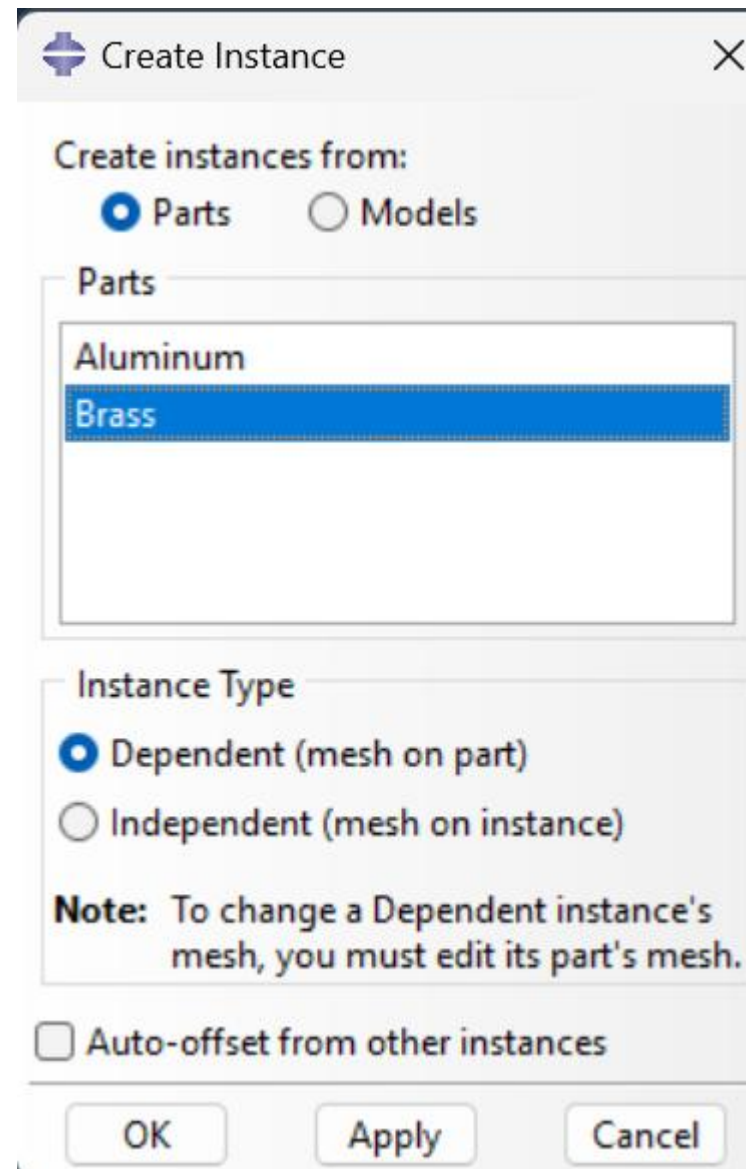
2. PROPERTY: ASSIGN PROPERTIES TO BRASS AND ALUMINUM PARTS

- Assign Section for Brass
 - Select Brass in “Part:”
 - Click on “Assign Section” icon
 - Select the part area of Brass rectangle
 - Click “Done”
 - In the “Edit Section Assignment”
 - Link “Section:” to “Brass”
- Assign Section for Aluminum
 - Select Aluminum in “Part:”
 - Click on “Assign Section” icon
 - Select the part area of Aluminum rectangle
 - Click “Done”
 - In the “Edit Section Assignment”
 - Link “Section:” to “Aluminum”



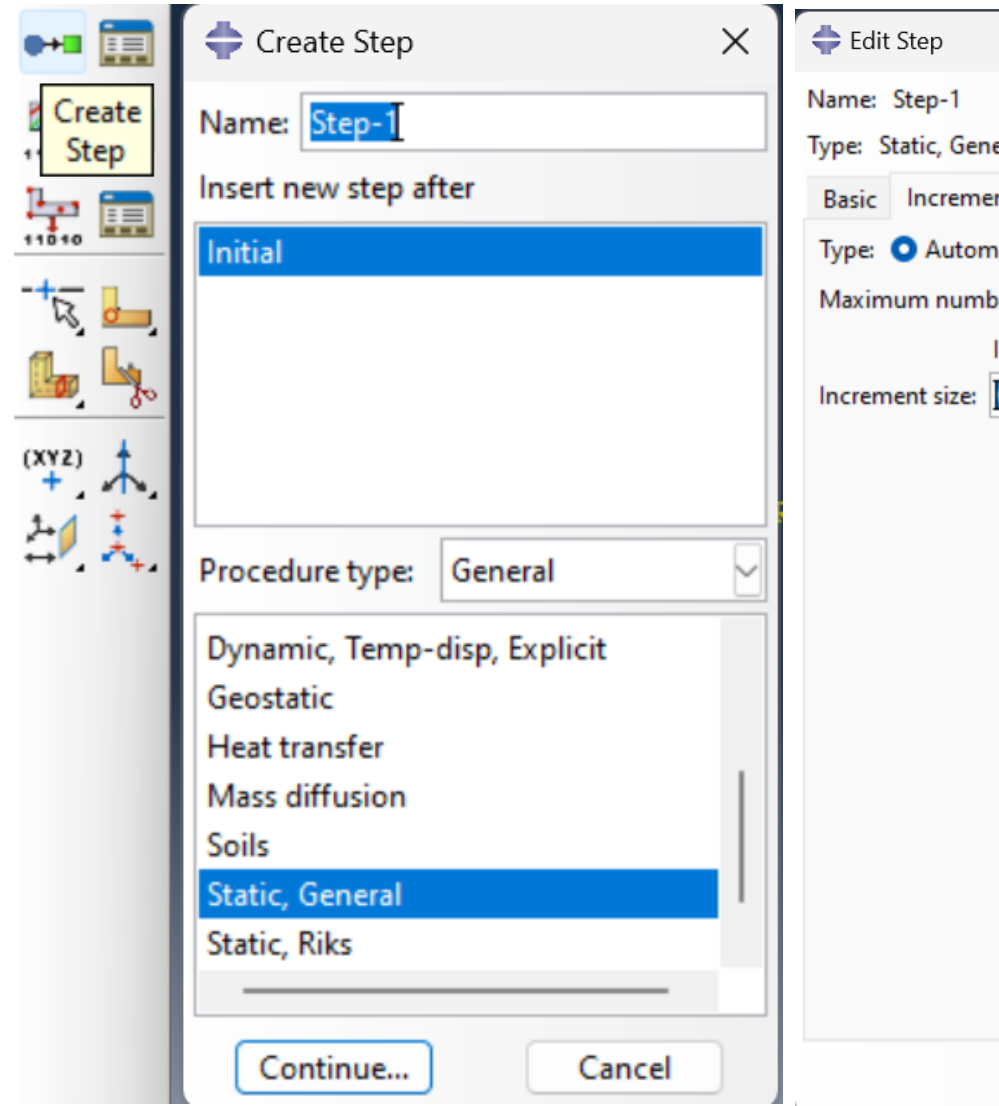
3. ASSEMBLY

- Create Instance
 - Tick “Parts” in “Create instances from:”
 - Double Click on “Brass” Part, then Double Click on “Aluminum” Part.
 - Select “Dependent (mesh on part)” as “Instance Type”
 - Click “Ok”
- Check number of your instances in “Model Database” to make sure there is no duplication of instances



4. STEP

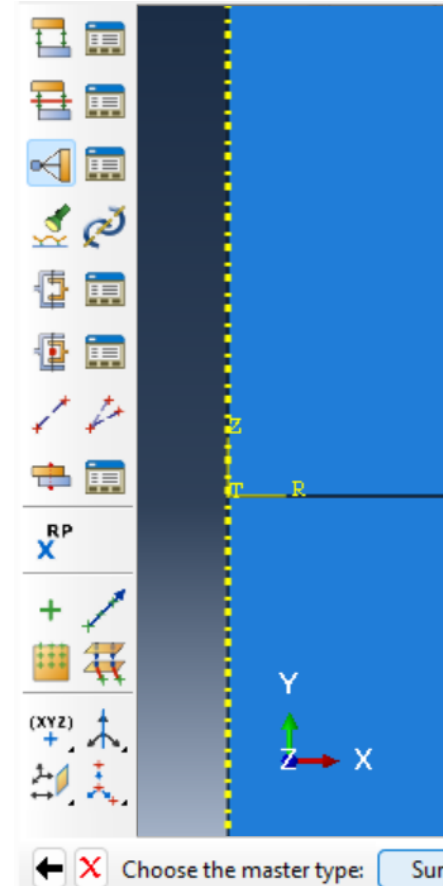
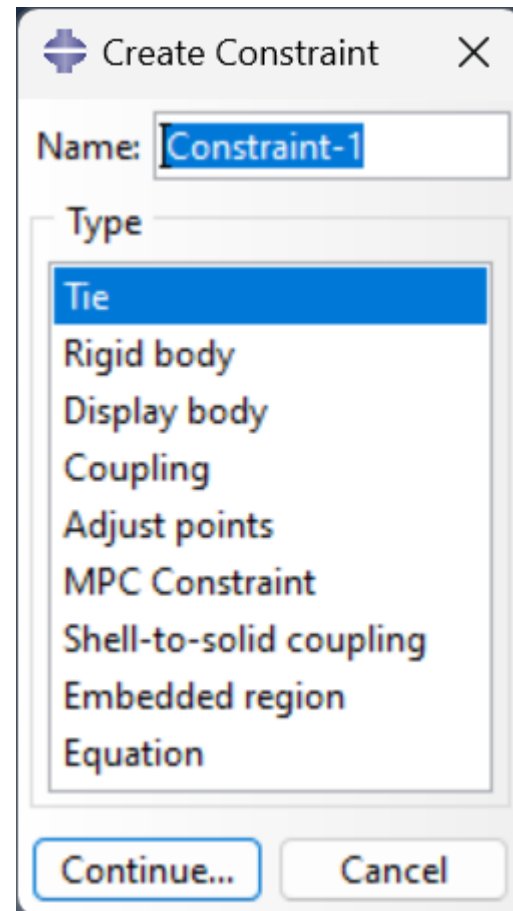
- Click on the “Create Step” icon
 - Input “Step-1” in “Name:”
 - Choose “Static, General” as “Procedure type:”
 - Click “Continue”
- In the “Edit Step” interface
 - Click on the “Incrementation” tab
 - Change the “Initial” of “Incrementation size:” to be “0.25”.
 - Click “Ok”



5. INTERACTION

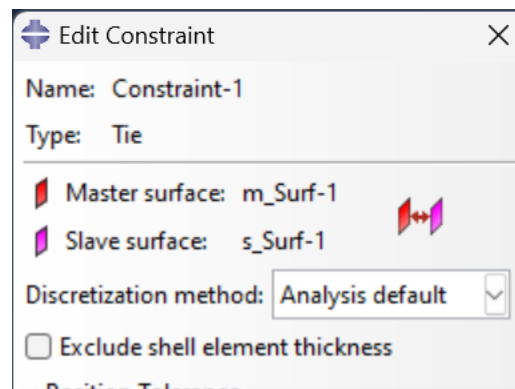
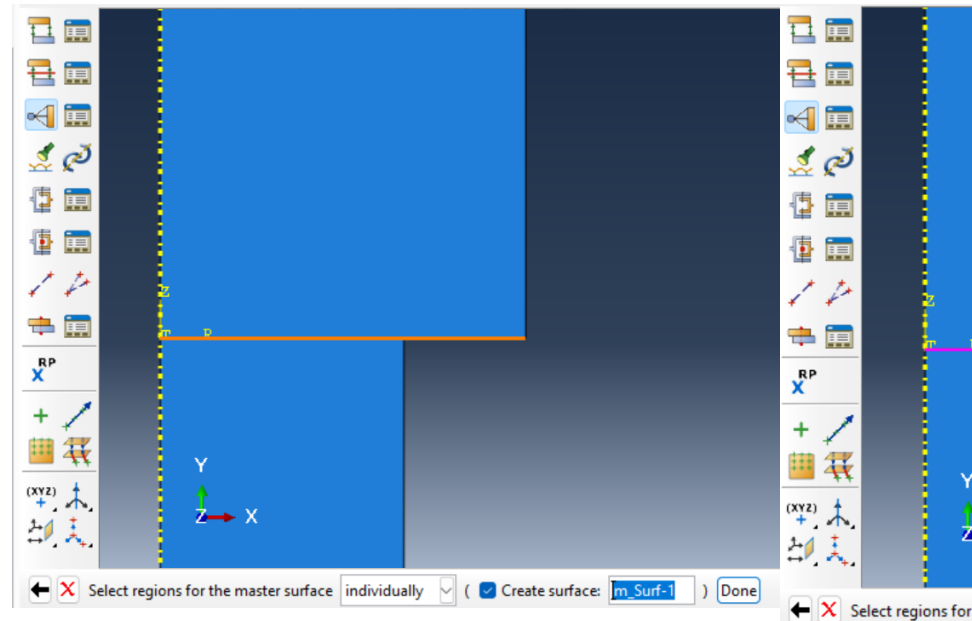
➤ Click on the “Create Constraint” icon

- Name your constraint in “Name:”
- Choose “Tie” as constraint “Type”
- Click “Continue”
- Choose “Surface” as the “master type:”



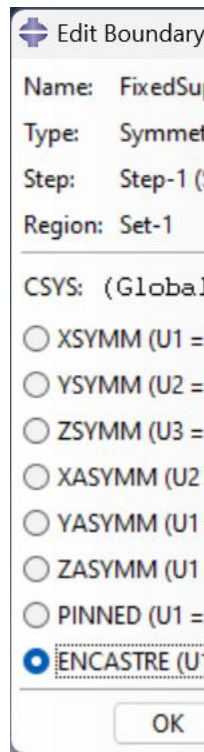
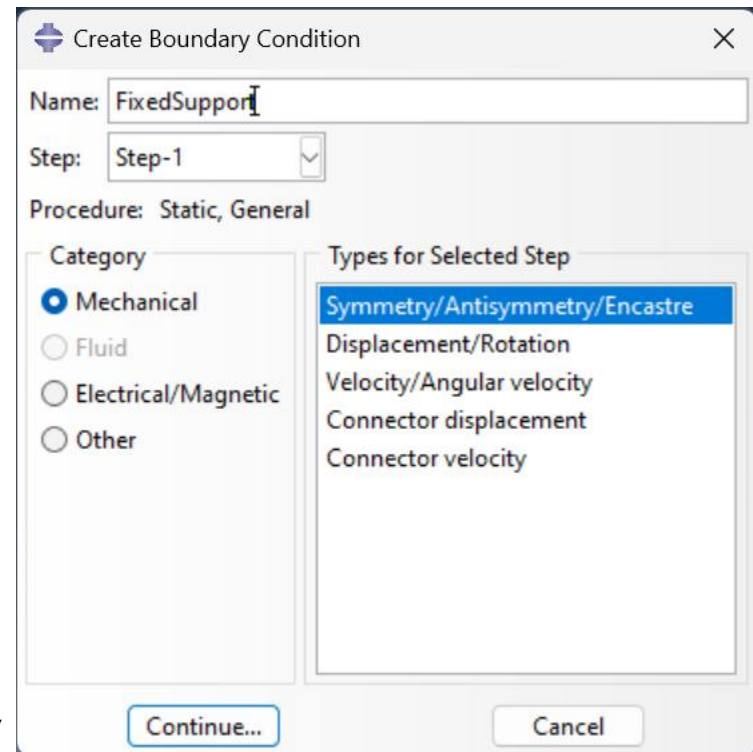
5. INTERACTION

- Click on the “Create Constraint” icon
 - Name your constraint in “Name:”
 - Choose “Tie” as constraint “Type”
 - Click “Continue”
 - Choose “Surface” as the “master type:”
 - Select regions for the master surface
 - Click “Done”
 - Select Surface as the “slave type:”
 - Select regions for the slave surface
 - Click “Done”
 - Click “Ok” in the “Edit Constraint” interface.



6. LOAD: SUPPORTS

- Click the “Create Boundary Condition” icon
 - Type “FixedSupport” as “Name:”
 - Select “Step-1” from “Step:”
 - Tick “Mechanical” as “Category”
 - Select “Symmetry/Antisymmetry/Encastre” as “Types for Selected Step”
 - Click “Continue”
 - Select the top line and the bottom line of the Assembly following “Select regions for the boundary condition”
 - Click “Done”
 - Tick “ENCASTRE” from the “Edit Boundary Condition” interface.
 - Click “Ok”



6. LOAD: TEMPERATURE

- Click the “Create Predefined Field” icon
 - Type “Temperature” as “Name:”
 - Select “Step-1” from “Step:”
 - Tick “Other” as “Category”
 - Select “Temperature” as “Types for Selected Step”
 - Click “Continue”
 - Select the areas of both Brass and Aluminum within the Assembly following “Select regions for the field”
 - Click “Done”
 - Input “42” in “Magnitude” from the “Edit Predefined Field” interface.
 - Click “Ok”

Create Predefined Field

Name: Predefined Field-1

Step: Step-1

Procedure: Static, General

Category

- ☐ Mechanical
- ☐ Fluid
- ☒ Other

Types for Selected Step

Temperature

Continue... Cancel

Edit Predefined Field

Name: Predefined Field-1

Type: Temperature

Step: Step-1 (Static, General)

Region: Set-2

Distribution: Direct

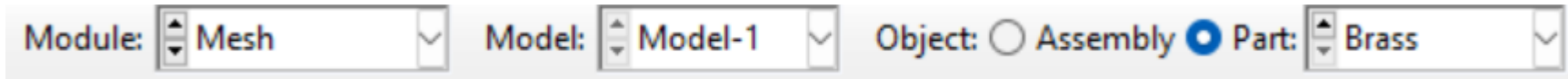
Section variation: Const

Magnitude: 42

Amplitude: (Ram)

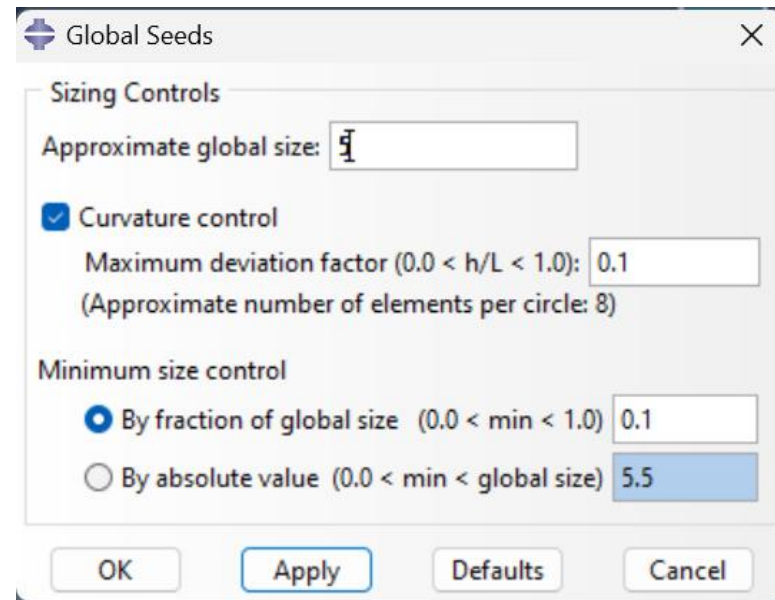
OK

7. MESH



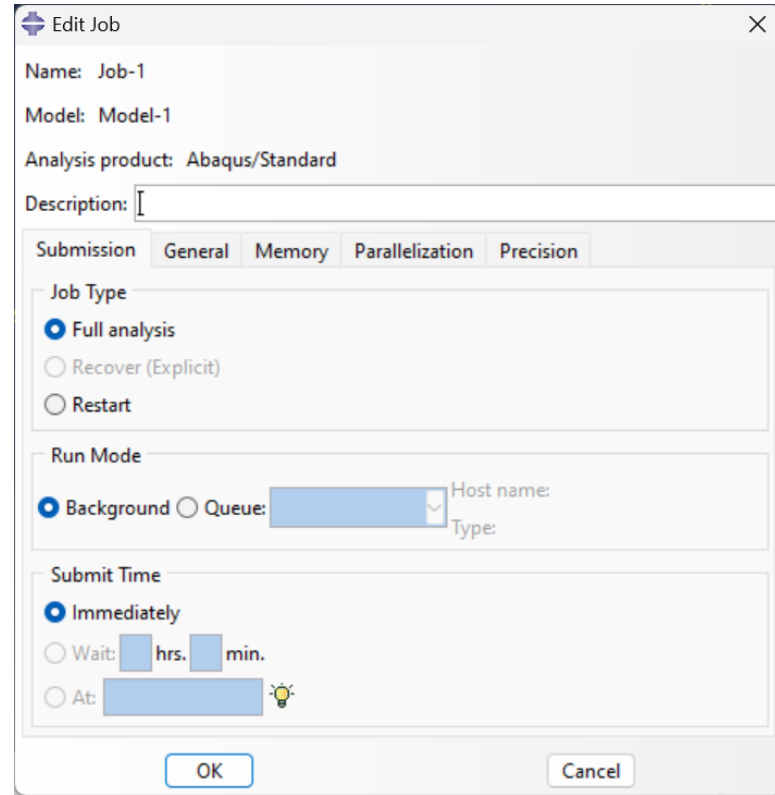
* Must do the meshing in Part because our Assembly is dependent

- Tick “Part” as “Object:”
- Select “Brass” as “Parts:”
- Click on “Seed Part” icon
 - Input numerical value, say 5 (**the finer the better but limited for Education Edition**), as “Approximate global size:”
 - Click “Ok”
- Click on “Mesh Part” icon
 - Click on “Yes” following “Ok to mesh the part?”
- Redo it again for “Aluminum” Part



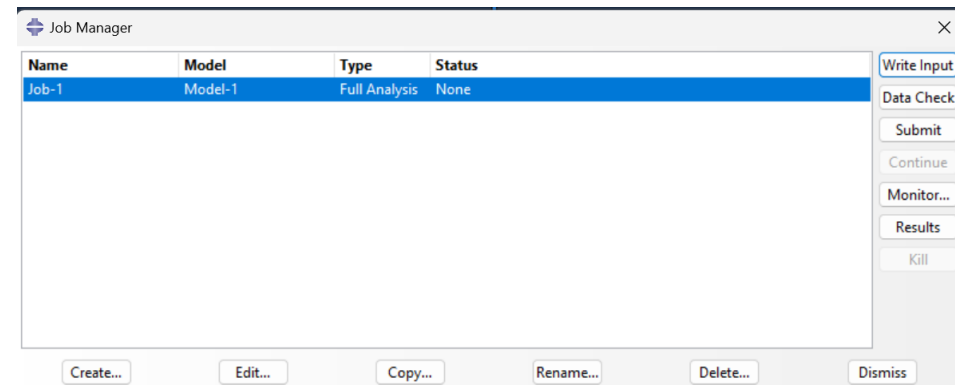
8. JOB

- Click on the “Create Job” icon
 - Name your job in “Name:”
 - Click “Continue”
 - Use all the default settings in the “Edit Job” interface
 - Click “Ok”
- Click on the “Job Manager” icon
 - Click on “Data Check”, wait until it is successful
 - Click “Submit”, wait until it is successful
 - Check the “Results” (S: Stress, E: Strain, U: Displacement)



The "Edit Job" dialog box is shown with the following settings:

- Name: Job-1
- Model: Model-1
- Analysis product: Abaqus/Standard
- Description: [Empty text box]
- Submission tab is selected.
- Job Type: ☒ Full analysis, ☐ Recover (Explicit), ☐ Restart
- Run Mode: ☒ Background, ☐ Queue: [Dropdown menu], Host name: [Text box], Type: [Text box]
- Submit Time: ☒ Immediately, ☐ Wait: [Text box] hrs. [Text box] min., ☐ At: [Text box] [Lightbulb icon]
- Buttons: OK, Cancel



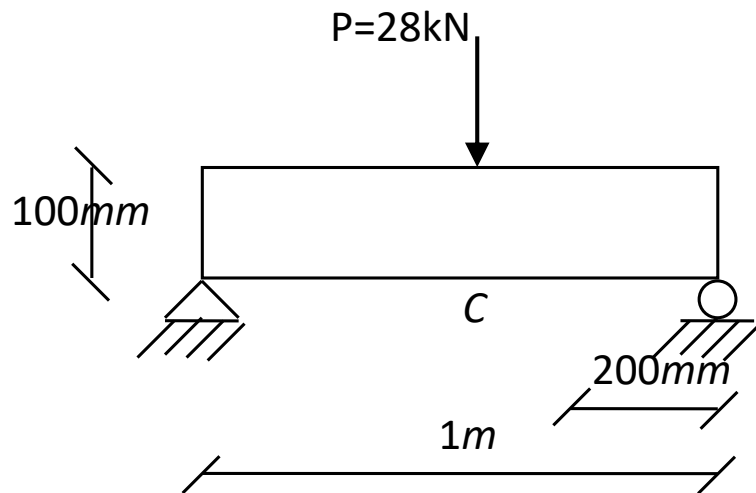
The "Job Manager" window displays a table of jobs and a sidebar with action buttons.

Name	Model	Type	Status
Job-1	Model-1	Full Analysis	None

Buttons on the right sidebar: Write Input, Data Check, Submit, Continue, Monitor..., Results, Kill.

Buttons at the bottom: Create..., Edit..., Copy..., Rename..., Delete..., Dismiss.

EXERCISE 2: 2D PLANAR



Width of the beam = 25mm

Material Properties:

$E = 200\text{ GPa}$, $\nu = 0.3$

$\sigma_{\text{allowable}} = 200\text{ MPa}$

1. PART

- Click on the “Create Part” icon
 - Type the “Name:”
 - Tick “2D Planar” in the “Modeling Space”
 - Keep others as default settings
 - Type 2000 in “Approximate size:”
 - Click “Continue”
- Draw the 2D plate as in the exercise (Better to construct the **rectangle by line**. **Middle points at the top and bottom** must be included)

Create Part

Name: 2DPlate

Modeling Space

☐ 3D ☒ 2D Planar ☐ Axisymmetric

Type

☒ Deformable
☐ Discrete rigid
☐ Analytical rigid
☐ Eulerian

Options

None available

Base Feature

☒ Shell
☐ Wire
☐ Point

Approximate size: 600

Continue... Cancel

2. PROPERTY

- Click on the “Create Material” icon
 - Type “Steel” in “Name:”
 - Select “Mechanical” tab, “Elasticity”, “Elastic” as “Material Behaviors”
 - Input “200e3” for “Young’s Modulus” and “0.3” as “Poisson’s Ratio”
 - Click “Ok”
- Click on the “Create Section” icon
 - Type “Steel” in “Name:”
 - Keep others as default
 - Click Continue
 - In the “Edit Section” interface, select “Steel” as “Material”, tick “Plane stress/strain thickness:”, Put the value “25”
- (Continue)

Edit Material

Name: Steel

Description:

Material Behaviors

Elastic

General Mechanical Thermal Electrical/Magnetic Other

Elastic

Type: Isotropic

☐ Use temperature-dependent data

Number of field variables: 0

Moduli time scale (for viscoelasticity): Long-term

☐ No compression

☐ No tension

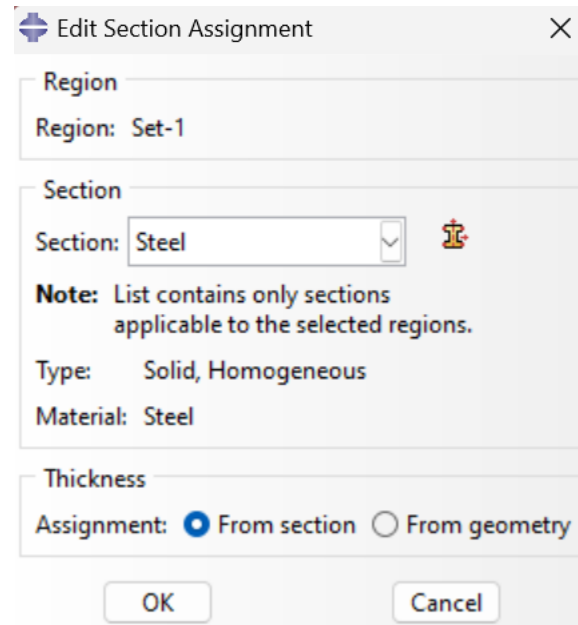
Data

	Young's Modulus	Poisson's Ratio
1	200e3	0.3

OK Cancel

2. PROPERTY

- Click on the “Assign Section” icon
 - Select the plate area to be assigned the Section. Click “Done”
 - In “Edit Section Assignment”, select “Steel” in “Section:”
 - Keep others as default settings
 - Click “Ok”



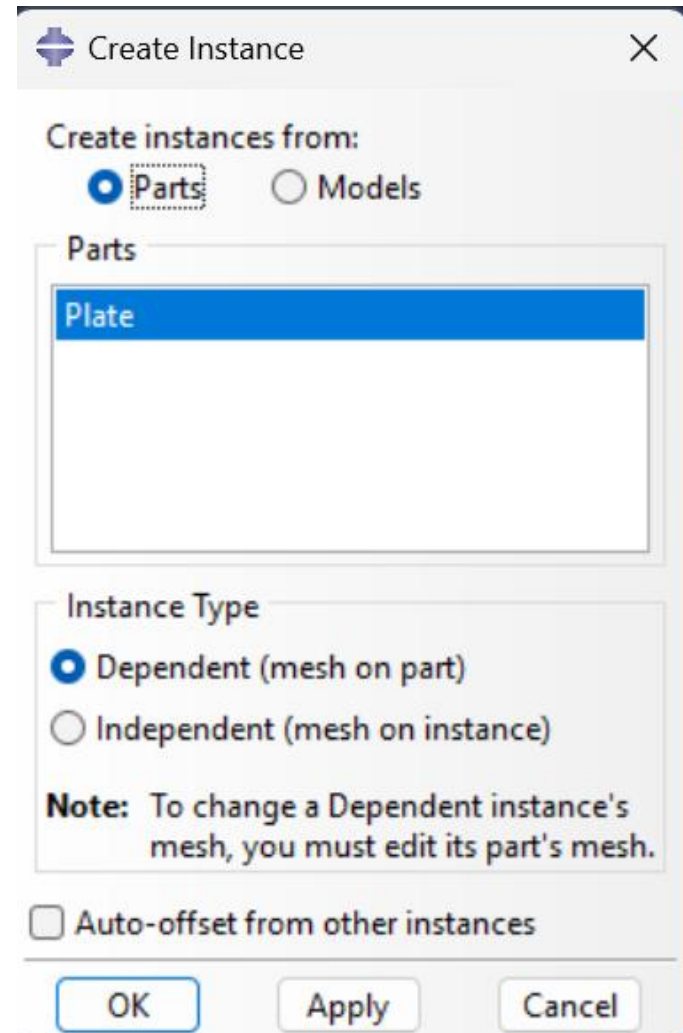
The screenshot shows the 'Edit Section Assignment' dialog box with the following settings:

- Region:** Set-1
- Section:** Steel (selected in a dropdown menu, with a small icon to the right)
- Note:** List contains only sections applicable to the selected regions.
- Type:** Solid, Homogeneous
- Material:** Steel
- Thickness:**
 - Assignment:** ☒ From section ☐ From geometry

Buttons at the bottom: OK and Cancel.

3. ASSEMBLY

- Click on the “Create Assembly” icon
- Tick “Parts” in “Create instances from:”
- Double Click on the part you want to create an Assembly.
- Select “Dependent (mesh on part)” as “Instance Type”
- Click “Ok”



4. STEP

- Same procedure as in Exercise 1.

5. INTERACTION

➤ Not necessary in this exercise

6. LOAD

- Create the Fixed Support
- Apply the concentrated force downward $F_2 = -28\text{e}3 \text{ N}$ at the middle and top of the beam.
- You need to create a point there when creating part.

7. MESH

- Use “Seed Edges” for the hole and the fillet zones to get the finest mesh as possible
- Use “Seed Part” to mesh other area. Not necessarily as fine as the hole and the fillet zone.
- Finally, use “Mesh Part Instance” to do the meshing

8. JOB

- Same procedure as in Exercise 1.